

Momentum strategies overweight assets that have recently risen in price and underweight assets that have fallen in price.¹⁸ Momentum is pervasive and observed in equities, fixed income, foreign exchange, real estate, commodities (where it is often called CTA, for the type of fund which often pursues it), within and across asset classes, and in asset classes all over the world.¹⁹ Similarly, selling volatility protection can be done in options markets where the underlying securities are equities, fixed income, commodities, or foreign exchange. An investor should decide on whether momentum or volatility strategies are appropriate and then implement them with appropriate asset classes in the cheapest way possible.

3.3. CANADA PENSION PLAN INVESTMENT BOARD

CPPIB, the fund manager of the Canada Pension Plan, uses two simple factors—stocks and bonds.²⁰ CPPIB holds many other asset classes—it is chock full of private equity, infrastructure and other sexy, illiquid assets, but it does not consider them to be asset classes. CPPIB looks through “private equity” and other labels and treats these investments as a combination of its two factors, stocks and bonds.

Figure 14.5, Panels A and B show factor investing à la CPPIB (which it calls the “total portfolio approach”) and are drawn from CPPIB’s 2013 annual report. Consider a private equity (buyout) investment in Panel A. CPPIB does not have a separate allocation to private equity and theoretically could hold almost all its portfolio in private equity subject to staying within the overall risk limits for the fund. In this particular example, \$1 invested in the private equity fund is economically equivalent to \$1.30 invested in public equities and a short position in bonds. This private equity investment is levered, and CPPIB accounts for this by the –\$0.30 position in fixed income. In reality, CPPIB performs a more complicated matching procedure that involves pairing the private equity fund by sector and geographic region, with adjustments for the artificially low volatility of reported private equity returns (see chapter 11).

Panel B illustrates a factor investing decision for real estate. In this example, \$1 invested in the real estate deal is economically equivalent to \$0.40 in public equities and \$0.60 of debt. Again, CPPIB’s process is a bit more complicated: it assigns real estate to one of three risk classes (low, core, or high), and does some

¹⁸ This is a *cross-sectional* strategy that takes positions in different assets at a point in time. This is different from *time-series* momentum, which refers to the tendency of a particular asset to keep rising if it has had past high returns.

¹⁹ An exception is that momentum is very weak in over-the-counter equities markets, as Ang, Shtaubert, and Tetlock show (2013).

²⁰ This is drawn from the case “Factor Investing: The Reference Portfolio and Canada Pension Plan Investment Board,” CaseWorks ID #120302.